

Unit 6: Cubic & Cube Root Functions

Mixed Transformations Practice

Name: _____

Period: _____

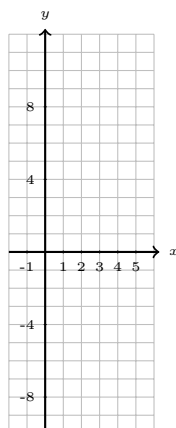
For each: Find a , h , k → Find center point → Transform anchor points → Graph → Find y -intercept

1. $f(x) = (x - 2)^3 + 1$

$a = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$

Center: ($\hspace{0.5cm}$, $\hspace{0.5cm}$)

Parent	New
$(-2, -8)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(-1, -1)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(0, 0)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(1, 1)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(2, 8)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)



y -int: _____

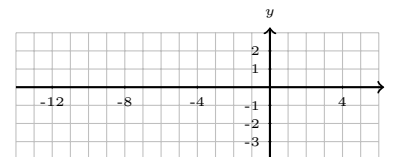
2. $f(x) = \sqrt[3]{x + 4} - 1$

$a = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$

Center: ($\hspace{0.5cm}$, $\hspace{0.5cm}$)

Parent	New
$(-8, -2)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(-1, -1)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(0, 0)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(1, 1)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(8, 2)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)

y -int: _____

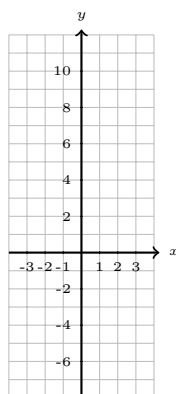


3. $f(x) = -x^3 + 2$

$a = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$

Center: ($\hspace{0.5cm}$, $\hspace{0.5cm}$)

Parent	New
$(-2, -8)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(-1, -1)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(0, 0)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(1, 1)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(2, 8)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)



y -int: _____

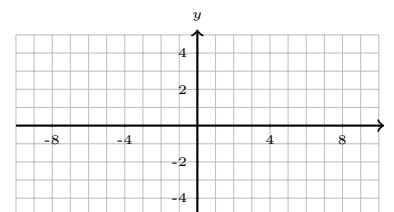
4. $f(x) = 2\sqrt[3]{x}$

$a = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$

Center: ($\hspace{0.5cm}$, $\hspace{0.5cm}$)

Parent	New
$(-8, -2)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(-1, -1)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(0, 0)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(1, 1)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(8, 2)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)

y -int: _____

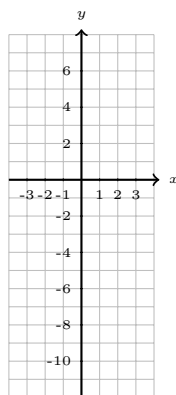


5. $f(x) = -(x + 1)^3 - 2$

$a = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$

Center: ($\hspace{0.5cm}$, $\hspace{0.5cm}$)

Parent	New
$(-2, -8)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(-1, -1)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(0, 0)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(1, 1)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(2, 8)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)



y -int: _____

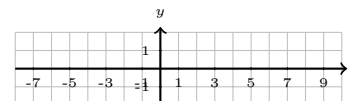
6. $f(x) = \frac{1}{2}\sqrt[3]{x - 1}$

$a = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$

Center: ($\hspace{0.5cm}$, $\hspace{0.5cm}$)

Parent	New
$(-8, -2)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(-1, -1)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(0, 0)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(1, 1)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)
$(8, 2)$	($\hspace{0.5cm}$, $\hspace{0.5cm}$)

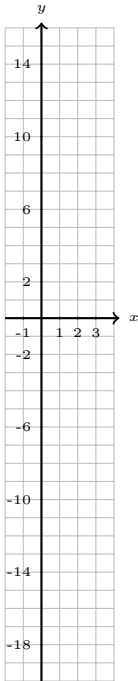
y -int: _____



7. $f(x) = 2(x - 1)^3 - 3$

$a = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$
Center: (,)

Parent	New
$(-2, -8)$	(,)
$(-1, -1)$	(,)
$(0, 0)$	(,)
$(1, 1)$	(,)
$(2, 8)$	(,)



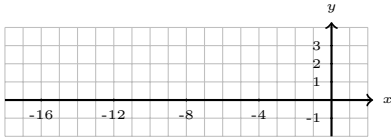
y -int: _____

8. $f(x) = -\sqrt[3]{x + 8} + 1$

$a = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$
Center: (,)

Parent	New
$(-8, -2)$	(,)
$(-1, -1)$	(,)
$(0, 0)$	(,)
$(1, 1)$	(,)
$(8, 2)$	(,)

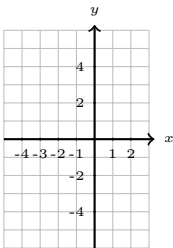
y -int: _____



9. $f(x) = \frac{1}{2}(x + 2)^3$

$a = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$
Center: (,)

Parent	New
$(-2, -8)$	(,)
$(-1, -1)$	(,)
$(0, 0)$	(,)
$(1, 1)$	(,)
$(2, 8)$	(,)



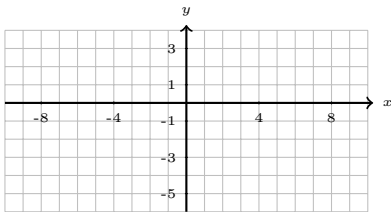
y -int: _____

10. $f(x) = -2\sqrt[3]{x} - 1$

$a = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$
Center: (,)

Parent	New
$(-8, -2)$	(,)
$(-1, -1)$	(,)
$(0, 0)$	(,)
$(1, 1)$	(,)
$(8, 2)$	(,)

y -int: _____



Write the equation from the description:

11. A cubic function that is reflected over the x -axis, shifted right 3, and shifted up 4.

$f(x) = \underline{\hspace{4cm}}$

12. A cube root function that is vertically stretched by a factor of 3, shifted left 2, and shifted down 5.

$f(x) = \underline{\hspace{4cm}}$